Condensed Matter Physics

Assignment - 2

1. Consider a one-dimensional diatomic lattice consisting of two types of atoms with masses m_1 and m_2 arranged alternately along a line with lattice spacing a. The atoms in the chain are connected to their nearest neighbour by springs with spring constants κ_1 and κ_2 as shown in Fig. 1:-

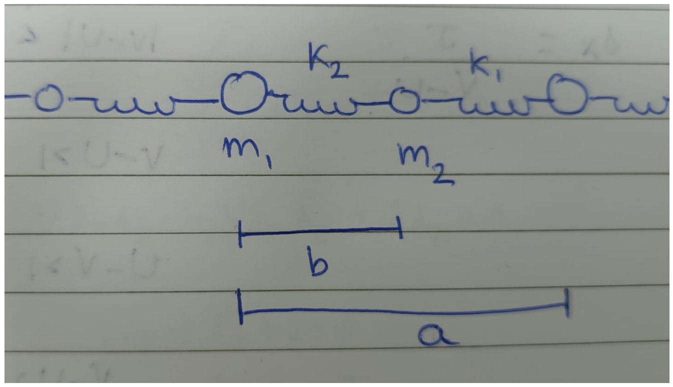

Figure 1: A diatomic chain of atoms with masses m_1 and m_2, with lattice spacing a and spring constant κ_1 and κ_2

   (a) Write down the total Hamiltonian of the system in terms of the displacement of the atoms from their equilibrium positions and the equations of motion of the displacements of the atoms with mass m_1 and m_2 in the $n-th$ unit cell of the diatomic chain.

   (b) Assume a plane wave form solution for the displacement of both the atoms and derive the dispersion relation $\omega(k)$ of the system.

   (c) Verify that in the limiting case of $m_1 = m_2$ and $\kappa_1 = \kappa_2$, the result reduces to the dispersion relation of the monoatomic chain.

   (d) Solve using the Fourier expansion and show that the dispersion consists of one gapped branch (optical) and one gapless branch (acoustic).

2. While performing the Sommerfeld expansion, we use the following relation

$$\int_0^\infty \frac{\eta^j}{e^\eta + 1} d\eta = \Gamma(j+1)(1-2^j)\zeta(j+1) \tag{1}$$

   The Riemann-Zeta function is given by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} \tag{2}$$

   Evaluate the above integral.

3. Consider a one-dimensional lattice with N sites and lattice constant a, where the single-particle dispersion relation for spinless electrons is

$$\epsilon(k) = -2t\cos(ka), \quad k \in \left[-\frac{\pi}{a}, \frac{\pi}{a}\right]. \tag{3}$$

   (a) The density of states g(E) of the system.

   (b) If the total number of spinless electrons is $N_E = \frac{N}{2}$, find out the Fermi energy E_F.

4. Consider a three-dimensional free electron gas at $T = 0$. In the absence of a magnetic field, both the spin-up and spin-down states are occupied up to the Fermi energy E_F. However, when a weak magnetic field B is applied, this degeneracy between spin-up and spin-down states is lifted.

   (a) Explain physically why only the electrons near the Fermi surface contribute to the magnetisation.

(b) Show that the imbalance in spin populations is proportional to the density of states at the Fermi level, $g(E_F)$, and to the field strength B.

(c) Using this reasoning, show that we can write the magnetisation as $M \propto g(E_F)\mu_B^2 B$.

5. Consider the single particle Hamiltonian in 1D in the presence of an artificial spin-orbit coupling which is given by,

$$\hat{H}_{1D} = \frac{1}{2m}(\hat{p}_x - \alpha\hat{\sigma}_z)^2 + \Omega\hat{\sigma}_x$$

where $\hat{\sigma_z}$ is the z-component of the usual Pauli spin matrix and Ω denotes Rabi coupling between the two internal spin states. Find out the eigenvalues of the above Hamiltonian and plot the corresponding dispersion relations.

6. A classical paramagnetic material consists of N non-interacting magnetic dipoles of magnitude μ at thermal equilibrium in a uniform external magnetic field B at temperature T. Find out the partition function, and from there obtain the magnetisation M. you can use the langevein function to derive,

$$L(x) = \coth(x) - \frac{1}{x},$$

. Find out the magnetic susceptibility

$$\chi = \left.\frac{\partial M}{\partial B}\right|_{B\to 0}.$$

at high temperature.

7. Consider a three-dimensional ideal Fermi gas at temperature $T \ll T_F$, T_F is the Fermi temperature. Starting from the general integral

$$I = \int_0^\infty H(E)\,\frac{1}{e^{(E-\mu)/k_BT}+1}\,dE,$$

, where H(E) is a polynomial function having the form $H(E) = E^\nu$, which vanishes at E=0. Using the saddle-point approximation, derive the Sommerfeld expansion up to order $(k_BT)^2$:

$$I \simeq \int_0^\mu H(E)\,dE + \frac{\pi^2}{6}(k_BT)^2 H'(\mu) + O(T^4).$$

. **Hints:**

- Use integration by parts and the substitution $x = (E-\mu)/k_BT$.
- Note that the derivative of the Fermi-Dirac distribution, $f'(E)$, is sharply peaked around $E = \mu$ at low temperatures.
- Expand $H(E)$ in a Taylor series around $E = \mu$ and retain terms up to $(E-\mu)^2$.

8. Consider a system of non-interacting spin-$\frac{1}{2}$ magnetic moments in a weak external magnetic field B. Show that to the leading-order, the magnetic susceptibility χ at high temperatures is given by **Curie's law**:

$$\chi = \frac{C}{T}, \tag{4}$$

and compute the **second-order correction** in $1/T$ to the susceptibility at high temperatures.